

Education

- **PhD in Physics**, [University of Illinois Urbana-Champaign](#) August 2020 – Expected August 2027
Advisor: Prof. Sascha Hilgenfeldt Urbana-Champaign, USA
- **MSc in Physics & PSI Certificate**, [Perimeter Institute](#) / [University of Waterloo](#) August 2018 – June 2019
Advisor: Prof. Freddy Cachazo Waterloo, Canada
- **MSc in Physics**, [Instituto de Física Teórica - UNESP](#) March 2018 – June 2020
Advisor: Prof. Ricardo Martínez-García São Paulo, Brazil
- **BSc in Physics**, [Pontificia Universidad Católica del Perú](#) March 2012 – December 2017
Advisor: Prof. José Bazo Lima, Peru
– Exchange Semester: [Ludwig-Maximilians-Universität München \(LMU Munich\)](#) April – September 2017
Munich, Germany

Awards & Fellowships

- Nominee for “Outstanding Staff and Student Contributors to Engineering Education Efforts”, [AE3 Engineering Education Community](#), University of Illinois Urbana-Champaign, 2025, 2026. Recognition for contributions to course enhancement and computational curriculum innovation as part of [SIIP](#).
- [Mavis Future Faculty Fellows Program](#), Grainger College of Engineering, UIUC, 2024 – 2025. A selective program for Ph.D. students aimed at training future faculty in research, teaching, and mentorship.
- [Department of Physics Graduate Student Travel Award Fund](#), UIUC, 2023, 2024 (two awards), 2025. Conferred to present research at the [Global Physics Summit 2025](#), [Boulder School 2024](#), [APS March 2024](#), and [APS March 2023](#).
- [SAIFR/IFT-Perimeter Fellowship](#) for outstanding results at the [Journeys into Theoretical Physics school](#). Included a full scholarship for [Perimeter Scholars International \(PSI\)](#), PI, 2018 – 2019, and fully funded master’s studies at [IFT-UNESP](#) by [CAPES](#) and [ICTP-SAIFR](#), 2018 – 2019.
- [Student Mobility Scholarship](#), conferred for exceptional academic performance to participate in international exchange, PUCP, 2017.
- [Program to Support Research Initiation \(PAIN\)](#), granted in group category by the project: “Experiments of Quantum Mechanics with individual photons: an instrument to complement undergraduate courses and for insertion into advanced research - Part II”, PUCP, 2015.

Publications

Research Papers

- **Jairo M. Rojas**, Mayisha Z. Nakib, and Sascha Hilgenfeldt. A Rational Tissue Vertex Model Based on Continuum Mechanics. *In preparation* (Expected Late 2026)
- **Jairo M. Rojas**, Naama Brenner, and Ahmed El Hady. Physics of Cellular Foraging: Resource Sensing, Nutrient-Dependent Decision Making and Population Movement. *In preparation* (Expected Late 2026)
- **Jairo M. Rojas**, Paul Robin, Frank Jülicher, and Ricard Alert. Capillary Interactions Organize Bacterial Colonies. *In preparation* (Expected Late 2026)
- W. L. Chang, **J. M. Rojas Huamaní**, G. K. Schaer, T. Bretl, S. Eggl, T. Golecki, M. Ornik, S. M. Salapaka, M. West, & S. Hilgenfeldt (2026). Computational Tools for Dynamics and Controls: Developing Computational Literacy for Engineering Students. In *Proceedings of the 2026 ASEE Annual Conference & Exposition*. DOI: [10.18260/1-2-59025](https://doi.org/10.18260/1-2-59025) [ASEE PEER](#)
- **Jairo M. Rojas**, Mayisha Z. Nakib, Vivian W. Tang, William M. Briehner, and Sascha Hilgenfeldt (2025). Cell Deformation Signatures along the Apical-Basal Axis: A 3D Continuum Mechanics Shell Model. DOI: [10.48550/arXiv.2501.17810](https://doi.org/10.48550/arXiv.2501.17810) [arXiv:2501.17810](#)
- Budzik, K., Rapčák, M., & **Rojas, J.** (2021). Conformal defects from string field theory. *Journal of High Energy Physics*, 2021(1), 120. DOI: [10.1007/JHEP01\(2021\)120](https://doi.org/10.1007/JHEP01(2021)120) [arXiv:2009.06567](#)
- Cachazo, F., & **Rojas, J.** (2020). Notes on biadjoint amplitudes, Tropic $G(3, 7)$ and $X(3, 7)$ scattering equations. *Journal of High Energy Physics*, 2020(4), 176. DOI: [10.1007/JHEP04\(2020\)176](https://doi.org/10.1007/JHEP04(2020)176) [arXiv:1906.05979](#)
- Adams, C., Del Tutto, M., Asaadi, J., Bernstein, M., Church, E., Guenette, R., **Rojas, J. M.**, Sullivan, H., & Tripathi, A. (2020). Enhancing neutrino event reconstruction with pixel-based 3D readout for liquid argon time projection chambers. *Journal of Instrumentation*, 15(04), P04009. DOI: [10.1088/1748-0221/15/04/P04009](https://doi.org/10.1088/1748-0221/15/04/P04009) [arXiv:1912.10133](#)

- Bazo, J., **Rojas, J.**, Best, S., Bruna, R., Endress, E., Mendoza, P., Poma, V., & Gago, A. (2018). Testing FLUKA on neutron activation of Si and Ge at nuclear research reactor using gamma spectroscopy. *Nuclear Instruments and Methods in Physics Research Section A*, 885, 1–6.
DOI: [10.1016/j.nima.2017.12.042](https://doi.org/10.1016/j.nima.2017.12.042) arXiv:1709.02026

Theses & Technical Reports

- **Rojas, J.** (2020). Aging as a Strategy to Prevent Extinction in Microbial Ecosystems. Master's Thesis (Advisor: Prof. Ricardo Martínez-García), *Instituto de Física Teórica - UNESP*. HDL: [11449/205021](https://hdl.handle.net/11449/205021)
- **J. Rojas**, R. Polifka, X. Poveda, & M. Moreno (2017). Boosted Decision Tree Optimization for the ATLAS search of ttH production in the 2ℓ same-sign channel. *CERN Document Server*. CDS: [2280193](https://cds.cern.ch/record/2280193)

Visiting Research Appointments

Moving Knowledge 2019

Harvard University, July – September 2019

Advisor: Prof. Roxanne Guenette

- Highly competitive international neutrino summer student program organized by Oxford and Harvard.
- Participation as a Research Scholar in the [Department of Physics at Harvard University](#).
- Collaboration with the Pixels/Wires [DUNE](#) group (resulting in publication [JINST 15 P04009](#)).

[CERN Summer Student Programme](#)

CERN, June – August 2017

Advisor: Dr. Richard Polifka

- Participation as a Non-Member State Summer Student at CERN in the [ATLAS EP-ADE-CA](#) group.
- Completed the six-week lecture programme from “Raw Data to Physics Results” to “Theoretical Concepts”.
- Research report [published on CERN Document Server](#).

Teaching & Mentoring Experience

Tutoring & Mentoring

- [Konstanz School of Collective Behaviour](#) (*Invited Tutor*) Summer 2026
Delivered two hands-on tutorials on *Modeling Active Matter*; mentored international student cohorts through their school research projects.
- Graduate Student Instructor, [Research Experience for Undergraduates \(REU\)](#) Program, UIUC 2024
Co-prepared and delivered a computational Wolfram Mathematica workshop for undergraduate research fellows.
- Graduate Student Mentor, [GPS UIUC](#) 2023 – 2026
- [Clubes de Ciencia Perú](#) (*Invited Tutor*) 2019
Collaborated with invited professors to set up hands-on experimental laboratories; guided participating students through experimental design and execution.

Pedagogical Training

- [Illinois Summer Teaching Institute \(ISTI\)](#), CITL, UIUC March 2022
Completed intensive pedagogical program *Amplifying Student Learning Across Modalities* hosted by the Center for Innovation in Teaching & Learning; trained in multimodal course design, innovative hybrid assessments, and inclusive student engagement strategies.

UIUC – Strategic Instructional Innovations Program (SIIP)

- [AI for Creative Computation](#) (*Student Developer*) Fall 2026 – Present
Developing instructional frameworks integrating generative AI with numerical simulation tools for 300-level dynamics and control courses (ME 340, ME 360, AE 352, AE 353).
- [Computational Tools for Dynamics & Control](#) (*Student Developer*) Spring 2024 – Spring 2026
Authored Python and Google Colab computational notebook modules for 300-level dynamics and control courses (ME 340, ME 360); strengthened computational literacy across engineering cohorts.

Teaching Assistantships

UIUC

- [TAM 541: Mathematical Methods I](#) Fall 2022
- [PHYS 325: Classical Mechanics I](#) Fall 2021
- [ME 310: Introductory Fluid Dynamics](#) Spring 2023
- [TAM 251: Introductory Solid Mechanics](#) Fall 2023
- [PHYS 214: University Physics: Quantum Physics](#) Spring 2022, Summer 2021
- [PHYS 213: University Physics: Thermal Physics](#) Spring 2022
- [PHYS 212: University Physics: Electricity & Magnetism](#) Fall 2020
- [PHYS 211: University Physics: Mechanics](#) Summer 2025, Spring 2021

PUCP

- FIS249: Classical Mechanics II Winter 2016
- FIS149: Physics III: Electricity & Magnetism Summer 2016

Leadership and Outreach Activities

- Service Director at [UIUC Peruvian Student Association](#). 2022 – 2023
- Assistant Staff at [Public Lecture Series](#), Perimeter Institute. 2019
- Member of the Organizing Committee of [Physics Colloquiums PUCP](#). 2015 – 2017
- Founding Vice President of the [Association of Students of Physics](#) at PUCP. 2016

Selected Conferences and Schools Attended

- [Junior Scientist Workshop on Theoretical Biophysics](#). Janelia Research Campus, USA, September 2026
Talk: Beyond Vertex Models: Elasticity Theory and Continuum Mechanics Insights into Epithelial Tissues.
Chalk Talk: Continuum Mechanics and the Constitutive Basis of Vertex Models.
- [WE-Heraeus Summer School: Active, Adaptive and Autonomous Matter](#). Cargèse, France, July 2026
- [Konstanz School of Collective Behaviour](#). Konstanz, Germany, July – August 2025
- [Joint March/April Meeting: Global Physics Summit 2025](#). Anaheim, USA, March 2025
Talk: [Linking cell mechanics and 3D cell morphology in confluent tissues](#).
- [Boulder School for Condensed Matter and Materials Physics](#). Boulder, USA, July 2024
Poster: [Beyond 2D Tissue Mechanics: A 3D Continuum Shell Theory for Epithelial Cells](#).
- [APS March Meeting 2024](#). Minneapolis, USA, March 2024
Talk: [Beyond 2D Tissue Modeling: A 3D Continuum Shell Theory of Epithelial Cells](#).
- [APS March Meeting 2023](#). Las Vegas, USA, March 2023
Talk: [Cell Deformation Signatures along the Apical-Basal Axis: A 3D Continuum Mechanics Shell Model](#).
- [4th Annual Conference on Quantitative Approaches in Biology](#). Evanston, USA, March 2022
Talk: Continuum Mechanics Modeling of In-plane Cell Anisotropy Induced by Out-of-plane Stresses.
- [2020 Wolfram Summer School](#). Online, June – July 2020
Talk: [Petri Nets Classification](#).
- [IFT - Perimeter-SAIFR Journeys into Theoretical Physics](#). São Paulo, Brazil, July 2016

Professional Memberships

- [American Physical Society \(APS\)](#) — [Division of Biological Physics \(DBIO\)](#), [Division of Soft Matter \(DSOFT\)](#), [Forum on International Physics \(FIP\)](#).

Technical & Computational Skills

- **Programming Languages:** Python, Julia, C++, Wolfram Language (Mathematica), Bash.
- **Research Computing & Infrastructure:** High-Performance Computing (Linux clusters, SLURM), Machine Learning, ROOT, FLUKA, Git.

Languages

Spanish	Native	
English	Advanced	TOEFL iBT: 112/120 & Cambridge FCE: B2
Portuguese	Advanced	UNESP - Exame de Língua Portuguesa: Passed
German	Intermediate	Goethe-Institut: Level B1 completed

Updated: September 26, 2026